

II B.E. (Computer Engineering)
Class Test III November 2023
3CERG1: Computer Organization and Architecture

Time : 70 Min.

Max. Marks 20

Note: Attempt as much as you can. Maximum marks awarded will not exceed 20.

- Q1** Describe all the steps of instruction execution with the help of instruction cycle state diagram. (4)
- Q.2** Let the memory locations num1 and num2, store values 30 and 50 respectively. The machine instruction LOAD store the value in Accumulator register (ACC). Let
Memory Address 30 contains value 10
Memory Address 50 contains value 20
What values do the following LOAD instructions store into accumulator?
(i) LOAD ACC, NUM1
(ii) LOAD ACC, (NUM2)
(iii) ADD ACC, NUM2 ; Assume ACC contains 30
(iv) ADD ACC, (NUM1) ; Assume ACC contains 20 (6)
- Q.3** Describe different variants of displacement addressing mode. (4)
- Q.4** Assume a pipeline with four stages: fetch instruction (FI), decode instruction (DI), fetch operand (FO), and execute (EX). Draw timing diagram for instruction pipeline for a sequence of 5 instructions, in which the third instruction is a branch instruction that takes to an address where remaining two instructions are executed. (6)
- Q.5** What is the difference between (i) a hardwired implementation and a microprogrammed implementation of a control unit? (ii) horizontal and vertical microinstructions? (6)
- Q6** Describe three different techniques for performing I/O operations. (4)